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SHELF ALGORITHMS FOR TWO-DIMENSIONAL PACKING PROBLEMS*

BRENDA S. BAKER† AND JERALD S. SCHWARZ‡

Abstract. This paper studies two approximation algorithms for packing rectangles, using the two-dimensional packing model of Baker, Coffman and Rivest [SIAM J. Comput., 9 (1980), pp. 846–855]. The algorithms studied are called next-fit and first-fit shelf algorithms, respectively. They differ from previous algorithms by packing the rectangles in the order given; the previous algorithms required sorting the rectangles by decreasing height or width before packing them, which is not possible in some applications. The shelf algorithms are a modification of the next-fit and first-fit decreasing height level algorithms of Coffman, Garey, Johnson and Tarjan [SIAM J. Comput., 9 (1980), pp. 808–826]. Each shelf algorithm takes a parameter r. It is shown that by choosing r appropriately, the asymptotic worst case performance of the shelf algorithms can be made arbitrarily close to that of the next-fit and first-fit level algorithms, without the restriction that items must be packed in order of decreasing height. Nonasymptotic worst case performance bounds are also investigated.

Key words. bin-packing, two-dimensional packing

1. Introduction. Two-dimensional packing problems arise in a variety of situations. For example, cutting-stock problems may involve cutting objects out of a sheet or roll of material so as to minimize waste. The scheduling of tasks with a shared resource involves two dimensions, the resource and time, and the problem is to pack (schedule) the tasks so as to minimize the total amount of time used. File management requires storing files in a storage medium, e.g., disk, while files may be created or destroyed over time. Baker, Coffman and Rivest [5] propose a combinatorial model for these situations. In this model, a rectangular bin with an open top (i.e., a semi-infinite dimension) is to be packed with a finite list of rectangles, of specified dimensions, such that the rectangles do not overlap and the total bin height used in the packing is as small as possible. The rectangles (also called "pieces") are to be packed with their sides parallel to the sides of the bin, and they may not be rotated. Such a packing is called an *orthogonal oriented* packing in the terminology of [5]. This model is illustrated in Fig. 1. For the cutting-stock problem, the width and height of the bin correspond to the width and length of the roll of material. For a computer scheduling application, where the shared resource is the main memory, the bin width corresponds to memory, while the semi-infinite height corresponds to time. This model is appropriate for studying the scheduling of tasks which share a resource such as memory, where each task requires a contiguous block of the resource. The scheduling of resources such as disk drives for which fragmentation is not an issue has been studied by Garey and Graham [8] using a fundamentally different multidimensional packing model.

For the Baker et al. model, the problem of determining an optimal packing of an arbitrary list of rectangles is NP-hard [5]. Therefore, Baker et al. [5], Golan [10] and Coffman et al. [7] studied polynomial time approximation algorithms for the problem. These algorithms all have the property that they sort the rectangles by decreasing height or width before packing them.

In many applications, the rectangles can be considered to form a queue which can be joined by new rectangles at any time. For example, in the computer scheduling

* Received by the editors January 29, 1979, and in final revised form July 9, 1982.
† Bell Laboratories, Murray Hill, New Jersey 07974.
‡ Bell Laboratories, Whippany, New Jersey 07981.

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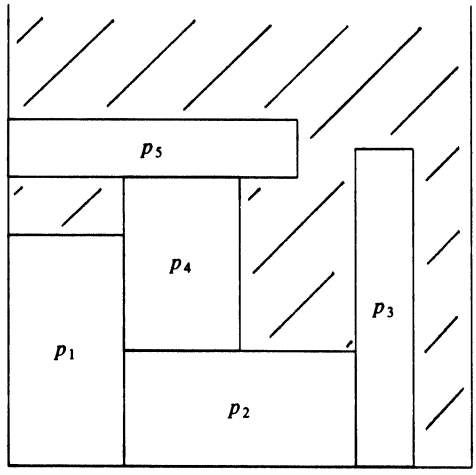

FIG. 1. *A possible packing of a list* $(p_1, p_2, p_3, p_4, p_5)$.

application with memory as the shared resource, a batch system could sort jobs by time or memory requirements before executing them, but an on-line system must handle jobs as they are created. In a warehouse, the boxes already there must be stored before new boxes arrive. In these situations it is not reasonable to wait for all the rectangles to enter the queue before packing them. Therefore, this paper studies polynomial time approximation algorithms (called *shelf* algorithms) which pack the rectangles in the order specified by the given list (queue), without sorting them first. Unlike the previous algorithms, the shelf algorithms can pack an infinite list of rectangles such that for any finite initial segment of the list, the height of the packing is within a constant times the height of an optimal packing of that segment.

The shelf algorithms are appropriate for situations such as the warehouse storage problem where (a) items to be packed are not all known in advance and (b) both dimensions represent resources fixed over time. The shelf algorithms will not handle applications in which one dimension is time, as in the computer scheduling application. Baker and Coffman [4] investigate algorithms for the situation where (a) items to be packed arrive over time, (b) one dimension is time and (c) preemption is allowed. The case in which items are not known in advance, one dimension is time and no preemption is allowed awaits further research.

Section 2 describes two kinds of shelf algorithms called next-fit and first-fit shelf algorithms. These are modifications of the next-fit decreasing height (NFDH) and first-fit decreasing height (FFDH) algorithms studied by Coffman et al. [7], but without the restriction of the previous algorithms that pieces be packed in order of decreasing height. Each shelf algorithm takes a parameter r, $0<r<1$, which is a measure of how much space is allowed in each shelf to handle variations in height of rectangles to come. The next-fit and first-fit algorithms with parameter r are called NFS_r and FFS_r, respectively.

Section 3 investigates the worst case bounds for NFS_r and FFS_r on lists of rectangles. The bounds obtained for the shelf algorithms fall into two classes: absolute worst case and asymptotic worst case. Let $A(L)$ denote the height used by algorithm A in packing list L, and let $\mathrm{OPT}(L)$ denote the height of an optimal packing of L. If α, β and H are constants such that for every list L of rectangles of height at most

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H, $A(L) \leqq \alpha$ OPT $(L) + \beta H$, then α is called an *asymptotic* worst case bound for A. If α is a constant such that for every list L of rectangles, $A(L) \leqq \alpha$ OPT (L), α is called an *absolute* worst case bound for A. The absolute worst case bound appears to be a better measure of performance when the number of rectangles to be packed is small, while the asymptotic bound is a better measure when the number of rectangles is large. Theorem 1 states that NFS$_r$ has a tight asymptotic worst case bound of $2/r$. Corollary 1.1 obtains a tight absolute worst case bound of $2/r + 1/r(1-r)$ for NFS$_r$. Corollary 1.2 states that this absolute worst case bound is minimized at $r \approx .634$, at which value the bound is approximately 7.46. The main theorem is Theorem 2, which states that FFS$_r$ has a tight asymptotic worst case bound of $1.7/r$. This theorem is important because it shows that we can get asymptotic performance arbitrarily close to the bound of 1.7 for FFDH [7], without the FFDH restriction that rectangles must be packed in order of decreasing height. Corollary 2.1 shows that $1.7/r + 1/r(1-r)$ is a tight absolute bound for FFS$_r$. Corollary 2.2 shows that this absolute worst case bound is minimal at $r \approx 622$, at which value it is approximately 6.99.

Section 4 investigates the worst case bounds for NFS$_r$ and FFS$_r$ on lists of squares. Theorem 3 proves a tight absolute worst case bound of $1/r + 1/r(1-r)$ for NFS$_r$ and FFS$_r$ on lists of squares. Note that this bound is lower than the absolute bound for arbitrary lists of rectangles. Corollary 3.1 shows that this absolute worst case bound for lists of squares is minimal at $r \approx .586$, at which value it is approximately 5.83. Proposition 1 and Theorems 4–6 investigate asymptotic worst case bounds on the NFS$_r$ and FFS$_r$ packings for squares. The bounds obtained are not tight, but they suggest that the asymptotic worst case performance is not significantly better for squares than for rectangles in general.

2. Shelf algorithms. Since the shelf algorithms are a modification of the next-fit and first-fit level algorithms, we begin by describing these.

The next-fit decreasing height (NFDH) level algorithm packs pieces in rows in order of decreasing height. The bottom of each row is called a *level*. The first level is the bottom of the bin. Pieces are packed from left to right, with their bottoms at the current level, until the next piece is too wide to fit. At this point, a new level is created along the horizontal line which coincides with the top of the tallest piece in the current level and packing continues at the new level. An NFDH packing is shown in Fig. 2a. For every list L of rectangles of height at most 1, NFDH $(L) \leqq 2$ OPT $(L) + 1$, and this bound is tight [7]. However, for any $k > 1$, there is a list L such that the height of the NFDH packing of L is at least k OPT (L) if the pieces are packed in the order given by L rather than in order of decreasing height (for a bin of width 1 and sufficiently small ε, let $L = (r_0, s_0, r_1, s_1, \cdots, r_k, s_k)$, where each r_i has height $1 + (i-k)\varepsilon$ and width ε, while each s_i has width 1 and height ε).

The first-fit decreasing height (FFDH) level algorithm is similar to the NFDH algorithm except that each piece is packed on the lowest level where it will fit, rather than on the top level; a new level is created when a piece won't fit on any existing level. An FFDH packing is shown in Fig. 2b. For every list L whose rectangles have height at most 1, FFDH $(L) \leqq 17/10$ OPT $(L) + 1$ [9]. However, if pieces are packed in the order given by L rather than by decreasing height, the performance can be arbitrarily bad compared to optimal.

In order to get algorithms which pack rectangles in a reasonable fashion without sorting first, we design our shelf algorithms to waste space purposely so that taller pieces arriving later can be accommodated. The name shelf algorithm is chosen by analogy with bookshelves, which can be filled in any order. Each shelf algorithm takes a parameter r, which is a measure of how much wasted space can occur. The shelf

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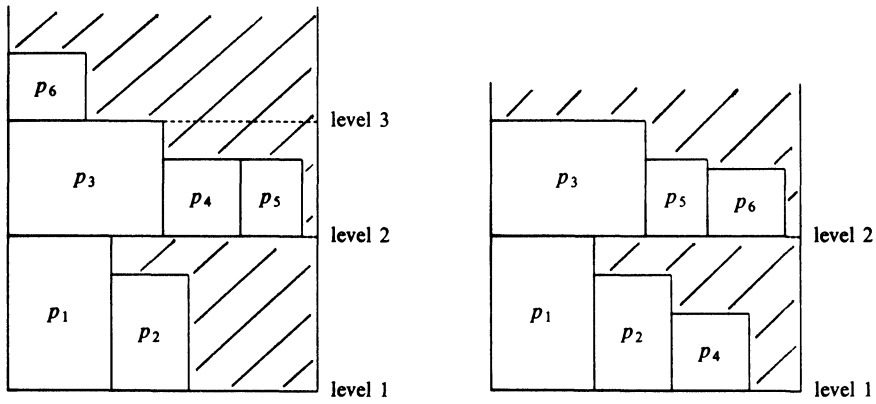
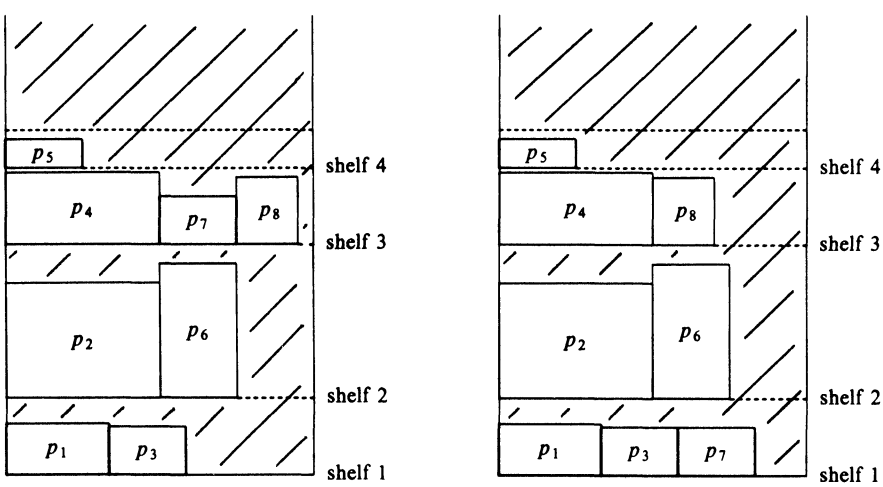


a) NFDH *packing* b) FFDH *packing*

FIG. 2. NFDH *and* FFDH *packings of a list* $L = (p_1, p_2, p_3, p_4, p_5, p_6)$.

algorithms pack pieces in rows (shelves), but each shelf is given a fixed height of r^k (for some integer k) when it is created. A rectangle of height h, $r^{k+1} < h \leqq r^k$, must be packed into a shelf of height r^k. The next-fit shelf algorithm with parameter r (denoted by NFS_r) packs each rectangle R as far left as possible into the top shelf of the appropriate height, if such a shelf exists and has room for R. Otherwise, a new shelf of the appropriate height is created on top of the top shelf in the bin, and R is placed at the left end of this shelf. An $\mathrm{NFS}_{.5}$ packing is shown in Fig. 3a. The first-fit shelf algorithm with parameter r (denoted by FFS_r) is similar except that each rectangle R is placed as far left as possible in the *lowest* shelf which is of the appropriate height and has room for R, if such a shelf exists, and into a new shelf of that height, otherwise. An $\mathrm{FFS}_{.5}$ packing is shown in Fig. 3b. It will be shown that as r approaches 1, the asymptotic performance of NFS_r and FFS_r improves. In fact, Theorem 2 shows that appropriate choice of r can bring FFS_r arbitrarily close to the asymptotic bound of



a) $\mathrm{NFS}_{.5}$ *packing* b) $\mathrm{FFS}_{.5}$ *packing*

FIG. 3. $\mathrm{NFS}_{.5}$ *and* $\mathrm{FFS}_{.5}$ *packings of a list* $L = (p_1, p_2, p_3, p_4, p_5, p_6, p_7, p_8)$.

17/10 obtained for the FFDH algorithm [7], even though the restriction of packing in order of decreasing height has been removed.

To simplify the proofs, it will be convenient to divide the shelves into two groups. The last shelf created for each height will be called *sparse*; the other shelves will be called *dense*. Let H_S denote the sum of the heights of the sparse shelves, and H_D the sum of the heights of the dense shelves.

3. Worst case bounds for lists of rectangles. This section investigates the asymptotic and absolute worst case bounds for NFS_r and FFS_r.

THEOREM 1. *For any constant r, $0<r<1$, and any list L whose rectangles have height at most H, $\mathrm{NFS}_r(L)<(2/r)\,\mathrm{OPT}(L)+H/r(1-r)$. Moreover, the asymptotic bound of $2/r$ is tight.*

Proof. We will show that $H_S<H/r(1-r)$ and that $H_D<(2/r)\,\mathrm{OPT}(L)$. Let the bin width be w.

First, we consider the sparse shelves. Let h be the size of the tallest sparse shelf. Within each shelf, the height of each piece is greater then r times the height of the shelf. Thus, $H>rh$ and

$$H_S<h\sum_{i=0}^{\infty} r^i=\frac{h}{1-r}<\frac{H}{r(1-r)}.$$

Now, we consider the dense shelves. For any dense shelf S, the next shelf S' of the same height as S contains a piece which would not fit in S. Consequently, the total width of the pieces in S and S' is at least w. Since each piece is taller than r times the height of its shelf, the total area of the pieces in S and S' is greater than $r/2$ times the area of S and S'. If we pair the shelves of each height in this manner (including the sparse shelf if the total number of dense shelves of that height is odd), we see that the total area of the rectangles (including the ones in the sparse shelves) is greater than $(rw/2)H_D$ and cannot be packed in a height less than $(r/2)H_D$. Consequently,

$$H_D<\frac{2\,\mathrm{OPT}(L)}{r}.$$

To show tightness, let the bin width be 1 and consider lists of the form $L=(r_1, s_1, r_2, \cdots, r_n, s_n)$, where $n>0$, each r_i has width $1/2-\Delta$ (for some Δ, $0<\Delta<1/3n$), each s_i has width 3Δ and all pieces have height $r+\varepsilon$ (for some ε, $0<\varepsilon<1-r$). The NFS_r packing places the r_i's and s_i's in pairs in shelves of height 1, using a total height of $n-1+r+\varepsilon$. An optimal packing is no worse than first packing the r_i's two to a row and then packing the s_i's in a single row on top for a total height of at most $(\lceil n/2\rceil+1)(r+\varepsilon)$. For $\alpha<2/r$ and β any constant, there exist sufficiently large n and sufficiently small ε that $n-1+r+\varepsilon>\alpha(\lceil n/2\rceil+1)(r+\varepsilon)+\beta(r+\varepsilon)$. □

Note that if $H=r^i$ for any integer i, the tallest shelf has height at most H rather than H/r, and the additive constant in Theorem 1 becomes $H/(1-r)$ rather than $H/r(1-r)$.

COROLLARY 1.1. *For any constant r, $0<r<1$, and any list L of rectangles,*

$$\mathrm{NFS}_r(L)<\left[\frac{2}{r}+\frac{1}{r(1-r)}\right]\mathrm{OPT}(L).$$

Moreover, this bound is tight.

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Proof. Since $H \leqq \mathrm{OPT}(L)$,

$$\mathrm{NFS}_r(L) < \frac{2}{r}\mathrm{OPT}(L) + \frac{H}{r(1-r)} \leqq \left[\frac{2}{r} + \frac{1}{r(1-r)}\right]\mathrm{OPT}(L)$$

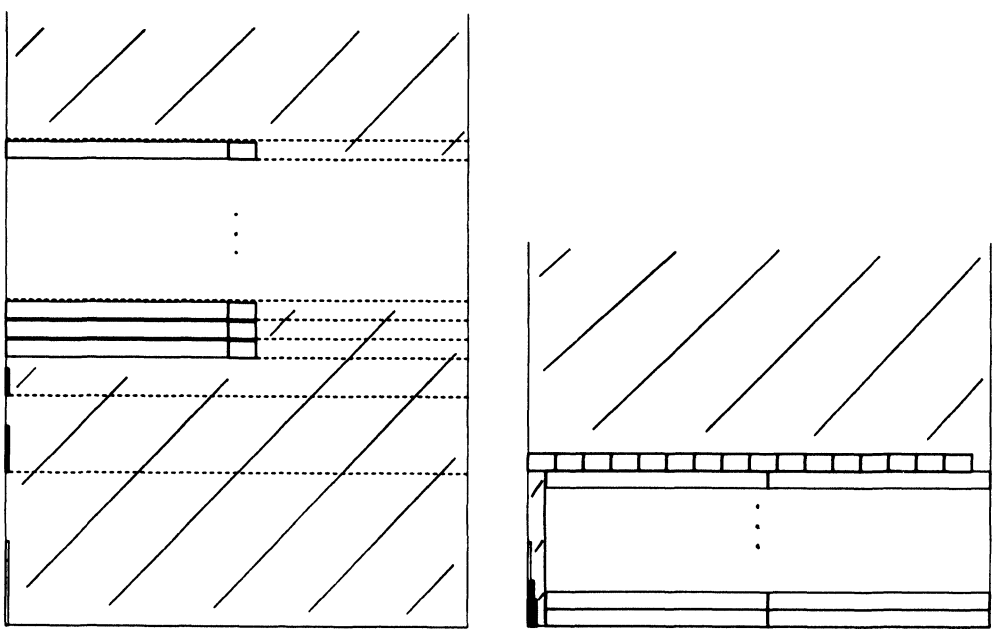
by Theorem 1. To show tightness, we will construct a list L such that in the NFS_r packing, the sparse shelves have a total height close to $1/r(1-r)$, the area of the dense shelves is close to $r/2$ times the area of the rectangles packed in them and the height of the optimal packing is close to the height of the tallest piece. Let the bin width be 1.

Let k be a positive integer. Let $n = 2\lceil r^{-k}\rceil$. Choose Δ and ε so that $3n\Delta < 1$ and $\varepsilon < r^k - r^{k+1}$. L will consist of pieces $(p_1, p_2, \cdots, p_k, t_1, s_1, t_2, s_2, \cdots, t_n, s_n)$, where each p_i has width Δ/k and height $r^i + \varepsilon$, each t_i has width $1/2 - \Delta$ and height $r^{k+1} + \varepsilon$ and each s_i has width 3Δ and height $r^{k+1} + \varepsilon$.

In the NFS_r packing, each p_i goes by itself in a shelf of height r^{i-1}, and the t_i's and s_i's are packed in pairs in shelves of height r^k (t_{i+1} will not fit into the shelf with t_i and s_i since their total width is $1/2 + 2\Delta$). The packing is shown in Fig. 4a. Thus,

$$\begin{aligned}\mathrm{NFS}_r(L) &\geqq \sum_{i=0}^{k-1} r^i + nr^k - (r^k - (r^{k+1} + \varepsilon)) \\ &= \frac{1}{1-r} - \frac{r^k}{1-r} + 2\lceil r^{-k}\rceil r^k - (r^k - (r^{k+1} + \varepsilon)) \\ &\geqq \frac{1}{1-r} - \frac{r^k}{1-r} + 2 - (r^k - r^{k+1}).\end{aligned}$$

An optimal packing of L is no worse than the following, depicted in Fig. 4b. Pack the p_i's at the bottom left of the bin, using width Δ and height $r + \varepsilon$. To the right



a) *The* $\mathrm{NFS}_{.5}$ *packing*

b) *An optimal packing*

FIG. 4. *Packings used in the proof of Corollary* 1.1 *with* $r = .5$, $k = 3$.

of the p_i's, pack the t_i's in rows of width $1-2\Delta$ and height $r^{k+1}+\varepsilon$. Since two of the t_i's fit in each row, the t_i's use at most height $(n/2)(r^{k+1}+\varepsilon)\leqq(1+r^{-k})(r^{k+1}+\varepsilon)\leqq r^{k+1}+r+(1+r^{-k})\varepsilon$. Finally, pack the s_i's in a row above the other pieces, using height $r^{k+1}-\varepsilon$. The total height used is at most $2r^{k+1}+r+(2+r^{-k})\varepsilon$. For any $\delta>0$, by taking k sufficiently large and ε sufficiently small, we get $\mathrm{NFS}_r(L)>[2/r+1/r(1-r)-\delta]\,\mathrm{OPT}(L)$. □

It is interesting to note that the asymptotic performance of NFS_r is nearly as good as that of NFDH, which has a tight asymptotic bound of 2 [7]. In fact, r can be chosen to bring the asymptotic performance as close as desired to 2, without the NFDH restriction that pieces be packed in order of decreasing height. Unfortunately, the absolute worst case performance gets worse as the asymptotic performance gets better. However, absolute worst case performance is crucial only for small numbers of pieces, and the shelf algorithms are intended for packings of large numbers of pieces.

Since the bound of Corollary 1.1 is tight, we can find the value of r with the best absolute worst case performance by differentiating the expression for the bound to find its minimum.

COROLLARY 1.2. *The value of r, $0<r<1$, for which the absolute worst case performance bound of* NFS_r *is minimal is* $(3-\sqrt{3})/2\approx.634$, *at which value* $\mathrm{NFS}_r(L)\approx 7.46\,\mathrm{OPT}(L)$ *for every list L of rectangles.*

One's intuition might suggest that the algorithm ought to do best at $r=1/2$. In fact, Theorem 2 shows that at $r=1/2$, $\mathrm{NFS}_r(L)$ can be arbitrarily close to 8 OPT (L), so that Corollary 1 gives a noticeable, if not large, improvement over $r=1/2$.

THEOREM 2. *For any r, $0<r<1$, and any list L of rectangles of height at most H,*

$$\mathrm{FFS}_r(L)<\frac{1.7}{r}\,\mathrm{OPT}(L)+\frac{H}{r(1-r)}.$$

Moreover, the asymptotic bound of $1.7/r$ *is tight.*

Proof. The proof is a modification of a proof from one-dimensional bin-packing. The one-dimensional bin-packing problem has a list L of numbers in the range (0, 1) which are to be packed in bins, each of which can hold a total of at most 1, and the goal is to minimize the number of bins used. This problem is equivalent to the two-dimensional problem with the restriction that all rectangles have the same height. In particular, the first-fit (FF) one-dimensional algorithm is equivalent to FFDH when all the rectangles have the same height. Garey, Graham, Johnson and Yao [9] showed that $\mathrm{FF}(L)\leqq 1.7\,\mathrm{OPT}(L)+1$ for every list L, where $\mathrm{FF}(L)$ and $\mathrm{OPT}(L)$ represent the number of bins used by FF and an optimal packing, respectively. We generalize the proof to handle shelves of different heights. Since the one-dimensional proof deals with bins which can hold at most 1, we assume without loss of generality that the bin width has been normalized to 1.

The original proof uses the following weighting function $W:[0,1]\to[0,8/5]$.

$$W(\alpha)=\begin{cases}\frac{6}{5}\alpha & \text{for } 0\leqq\alpha\leqq\frac{1}{6},\\ \frac{9}{5}\alpha-\frac{1}{10} & \text{for } \frac{1}{6}<\alpha\leqq\frac{1}{3},\\ \frac{6}{5}\alpha+\frac{1}{10} & \text{for } \frac{1}{3}<\alpha\leqq\frac{1}{2},\\ \frac{6}{5}\alpha+\frac{4}{10} & \text{for } \frac{1}{2}<\alpha\leqq 1.\end{cases}$$

The original proof showed that if $\bar{W}$ represents the sum of the weights of all pieces in L, then $\mathrm{FF}(L)\leqq\bar{W}+1$ and $\bar{W}\leqq 1.7\,\mathrm{OPT}(L)$. We extend the weighting idea of two dimensions by considering weighted areas. For each piece p, let $w(p)$ and $h(p)$

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represent the width and height of p, respectively. Let $\bar{A} = \sum_{p \in L} W(w(p))h(p)$. We will show that $\text{FFS}_r(L) < \bar{A}/r + H/r(1-r)$ and $\bar{A} \leqq 1.7\ \text{OPT}(L)$, and the result follows.

The original proof showed that for any packing of any list L, the sum of the weights of the pieces in any bin is at most 1.7 and deduced from this that $\bar{W} \leqq 1.7\ \text{OPT}(L)$. We can instead apply their proof along any horizontal line through the bin to find that the sum of $W(q_i)$ for the pieces q_i along this line is at most 1.7 and then integrate over the height of the optimal packing to obtain $\bar{A} \leqq 1.7\ \text{OPT}(L)$.

It remains to show that $\text{FFS}_r < \bar{A}/r + H/r(1-r)$. For $k \geqq 0$, let N_k be the number of shelves of height r^k. For $k \geqq 0$, let L_k be the sublist of L consisting of pieces whose heights are in the range $(r^{k+1}, r^k]$. If L'_k is the list of widths of pieces in L_k, then $\text{FF}(L'_k) = N_k$. By [9],

$$\sum_{p \in L_k} W(w(p)) \geqq \text{FF}(L'_k) - 1 = N_k - 1.$$

Since each piece is greater than r times the height of the shelf it is packed in,

$$\sum_{p \in L_k} W(w(p))h(p) \geqq r^{k+1}(N_k - 1).$$

Consequently, if the tallest shelf has height r^i,

$$\bar{A} \geqq \sum_{k=i}^{\infty} \sum_{p \in L_k} W(w(p))h(p) \geqq \sum_{k=i}^{\infty} r^{k+1}(N_k - 1)$$
$$= r\ \text{FFS}(L) - \frac{r^{i+1}}{(1-r)} > r\ \text{FFS}(L) - \frac{H}{(1-r)}$$

and

$$\text{FFS}(L) \leqq \frac{\bar{A}}{r} + \frac{H}{r(1-r)}.$$

Johnson et al. [11] proved that for every positive integer n, there is a list L for which $\text{OPT}(L) = n$ and the one-dimensional first-fit (FF) algorithm has $\text{FF}(L) \geqq 1.7n - 8$. If we turn L into a list of rectangles by letting each rectangle have height $H = r + \varepsilon$, where ε is sufficiently small, we get $\text{FFS}_r(L) \geqq 1.7n + r + \varepsilon - 9$ and $\text{OPT}(L) = n(r+\varepsilon)$. Thus, for every $\alpha < 1.7/r$ and every constant β, there are sufficiently large n and sufficiently small ε that $\text{FFS}_r(L) > \alpha\ \text{OPT}(L) + \beta H$. □

Note again that if $H = r^i$ for some integer i, the tallest shelf has height at most H and the additive constant in Theorem 2 becomes $H/(1-r)$ rather than $H/r(1-r)$.

COROLLARY 2.1. *For every constant r, $0 < r < 1$, and every list L of rectangles,*

$$\text{FFS}_r(L) < \left[\frac{1.7}{r} + \frac{1}{r(1-r)}\right] \text{OPT}(L).$$

Moreover this bound is tight.

Proof. Since $H \leqq \text{OPT}(L)$,

$$\text{FFS}_r(L) < \frac{1.7}{r}\text{OPT}(L) + \frac{H}{r(1-r)} \leqq \left[\frac{1.7}{r} + \frac{1}{r(1-r)}\right] \text{OPT}(L)$$

by Theorem 2. We know from [11] that for every r and every positive integer n, there exists a list L of rectangles, all of the same height d, such that the number of shelves used by FFS_r to pack L is at least $1.7n - 8$ while an optimal packing is no worse than placing the pieces in n rows each of height d. Let k and t be positive integers, $k < t$,

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and let $\varepsilon < r^{t-1} - r^t$. Let $n = \lceil r^{1-t} \rceil$. If we take $d = r^t + \varepsilon$, we have

$$\mathrm{FFS}_r\,(L) \geqq (1.7n - 9)r^{t-1} + r^t + \varepsilon, \qquad \mathrm{OPT}\,(L) = n(r^t + \varepsilon).$$

Let P_1 and P_2 be the FFS_r packing of L and an optimal packing of L, respectively. Let m be the maximum number of rectangles which appear in any single shelf in P_1. For some $x > 0$, the width of the empty area at the bottom right end of each shelf is smaller by at least x than every piece packed higher. Choose Δ so that $\Delta < x/(m+1)k$. Form a list L' from L by shrinking the width of each piece by $k\Delta$. If we replace each piece in P_1 by its corresponding piece in L', the total extra space in each shelf has width at most $mk\Delta < x$, so that no piece packed in a higher shelf will fit into a lower shelf. Therefore, $\mathrm{FFS}_r\,(L')$ uses the same number of shelves as $\mathrm{FFS}_r\,(L)$. In P_2, if we replace each piece by its corresponding piece in L' and push each piece as far left as possible, we get an empty region of width $k\Delta$ at the right end of the bin.

Now, let $L'' = (s_1, s_2, \cdots, s_k)$ where s_i has height $r^i + \varepsilon$ and width Δ. Let L_0 be a list consisting of the pieces of L' followed by the pieces of L''.

An optimal packing of L_0 is no worse than packing L'' into the bin using width $1 - k\Delta$ and height $n(r^t + \varepsilon)$, and then packing $s_1, s_2, \cdots, s_k$ left to right into the remaining space of width $k\Delta$. Thus, $\mathrm{OPT}\,(L_0) \leqq n(r^t + \varepsilon) = \lceil r^{1-t} \rceil (r^t + \varepsilon) \leqq (r^{1-t} + 1)(r^t + \varepsilon) \leqq r + r^t + r^{1-t}\varepsilon + \varepsilon$.

Now, FFS_r packs each s_i alone in a shelf of height r^{i-1}. These shelves use a total height of

$$\sum_{i=1}^{k} r^{i-1} = \frac{1}{1-r} - \frac{r^k}{1-r}.$$

The remaining shelves use a total height of at least $(1.7n - 9)r^{t-1} + r^t + \varepsilon$. Thus,

$$\begin{aligned}
\mathrm{FFS}_r\,(L_0) &\geqq \frac{1}{1-r} - \frac{r^k}{1-r} + (1.7n - 9)r^{t-1} + r^t + \varepsilon \\
&\geqq \frac{1}{1-r} - \frac{r^k}{1-r} + (1.7r^{1-t} - 9)r^{t-1} \\
&\geqq \frac{1}{1-r} - \frac{r^k}{1-r} + 1.7 - 9r^{t-1}.
\end{aligned}$$

For any $\delta > 0$ and sufficiently small ε and sufficiently large k and t, we obtain $\mathrm{FFS}_r\,(L_0) > [1.7/r + 1/r(1-r) - \delta]\,\mathrm{OPT}\,(L_0)$. □

We have seen that the asymptotic performance of FFS_r can be made as close as desired to 1.7, which is the asymptotic bound for FFDH [7], without the FFDH restriction of packing pieces in order of decreasing height. For the absolute worst case performance, we can find the minimal value of the bound of Corollary 2.1 by differentiating to get the following Corollary.

COROLLARY 2.2. *The value of r, $0 < r < 1$, for which the absolute worst case bound of FFS_r is minimal is $r \approx .622$. At this value of r, for every list L, $\mathrm{FFS}_r\,(L) \approx 6.99\,\mathrm{OPT}\,(L)$.*

4. Worst case bounds for lists of squares. Many two-dimensional packing algorithms have been found to have lower worst case bounds for lists of squares than for arbitrary lists of rectangles. For absolute worst case bounds, this pattern also holds for FFS_r and NFS_r.

THEOREM 3. *Let $0 < r < 1$. For any list L of squares, $\mathrm{NFS}_r\,(L) \leqq [1/r + 1/r(1-r)]\,\mathrm{OPT}\,(L)$ and $\mathrm{FFS}_r\,(L) \leqq [1/r + 1/r(1-r)]\,\mathrm{OPT}\,(L)$. Moreover, these bounds are tight.*

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Proof. Without loss of generality, let the bin width be 1. Let P be an NFS_r or FFS_r packing of a list L of squares, and let $|P|$ denote the height used by P. For $k \geqq 0$, let H_k denote the total height of shelves of height r^k and let A_k denote the total area of squares packed in shelves of height r^k. Suppose the height of the highest shelf is r^p. Define $h = r^p$. Let $A = \sum_k A_k$ be the total area of squares in L.

First, note that if $r^{k+1} > 1/2$, then every shelf of height r^k contains a square of size at least $r^{k+1} > 1/2$ and $A_k \geqq (r/2)H_k$.

Next, we claim that if there are $j > 1$ shelves of height r^k, then $A_k \geqq (r/2)H_k$. Note that the total width of squares packed in any two successive shelves of height r^k is at least 1 and each square on such a shelf has height at least r^{k+1}. Consequently, if j is even, $A_k \geqq (r/2)H_k$. So suppose j is odd. Pairing the first $j-3$ shelves of height r^k shows that the total area of squares in them is at least $(r/2)(H_k - 3r^k)$. If shelf $j-2$ is filled to width at least $1/2$, then this shelf plus the last pair of shelves contain squares of area at least $(r/2)(3r^k)$ and $A_k \geqq (r/2)H_k$. So suppose shelf $j-2$ is filled to width less than $1/2$. Then $r^{k+1} < 1/2$, shelf $j-1$ contains a square of size at least $1/2+\delta$, for some $\delta > 0$, and the total width of squares in shelves $j-1$ and j is at least 1. The total area of squares in these three shelves is at least

$$\begin{aligned}(\tfrac{1}{2}-\delta)r^{k+1} + (\tfrac{1}{2}+\delta)^2 + (1-(\tfrac{1}{2}+\delta))r^{k+1} &\geqq (\tfrac{1}{2}-\delta)r^{k+1} + (\tfrac{1}{2}+\delta)r^{k+1} + (\tfrac{1}{2}+\delta)\delta + (\tfrac{1}{2}-\delta)r^{k+1} \\ &\geqq (\tfrac{1}{2}-\delta)r^{k+1} + (r^{k+1}+\delta)\delta + r^{k+1} \\ &\geqq \tfrac{3}{2}r^{k+1}.\end{aligned}$$

Again, we have $A_k \geqq (r/2)H_k$.

Suppose that either there are $m > 1$ shelves of height $h = r^p$ or $rh > \frac{1}{2}$. From above, $A_p \geqq (r/2)H_p$. By pairing the shelves of other heights, we see that for $k > p$, $A_k \geqq (r/2)(H_k - r^k)$. Recall that the sums of the heights of the sparse and dense shelves are denoted by H_S and H_D, respectively. Consequently,

$$A = \sum_{k \geqq p} A_k \geqq \frac{r}{2} \sum_{k \geqq p} (H_k - r^k) + \frac{r}{2} h \geqq \frac{r}{2}(H_D + h).$$

Since $H_S \leqq \sum_{i=0}^{\infty} r^i h = h/(1-r)$, we have

$$\begin{aligned}|P| = H_S + H_D \leqq \frac{h}{1-r} + \frac{2A}{r} - h &= \frac{rh}{1-r} + \frac{2A}{r} \\ &\leqq \left[\frac{1}{1-r} + \frac{2}{r}\right] \mathrm{OPT}\,(L) \\ &= \left[\frac{1}{r} + \frac{1}{r(1-r)}\right] \mathrm{OPT}\,(L),\end{aligned}$$

which is the desired result.

So we may assume that there is only one shelf of height h and $rh < 1/2$. Define

$$\alpha = \frac{r(1-r)H_S}{\mathrm{OPT}\,(L)}.$$

Since $H_S \leqq h/(1-r) \leqq \mathrm{OPT}\,(L)/r(1-r)$, we know that $0 < \alpha \leqq 1$.

Note that each dense shelf has height at most rh and hence is filled to width at least $1-rh$. Moreover, the (sparse) shelf of height h contains a square of size at least

rh. Consequently,

$$\begin{aligned}\mathrm{OPT}(L) \geqq A &\geqq r(1-rh)H_D+(rh)^2\\ &\geqq r(1-rh)H_D+r^2h(1-r)H_S\\ &\geqq r(1-rh)H_D+rh\alpha\ \mathrm{OPT}(L),\end{aligned}$$

and

$$\frac{H_D}{\mathrm{OPT}(L)} \leqq \frac{1-\alpha rh}{r(1-rh)}.$$

Then

$$\begin{aligned}\frac{|P|}{\mathrm{OPT}(L)}=\frac{H_D+H_S}{\mathrm{OPT}(L)} &\leqq \frac{1-\alpha rh}{r(1-rh)}+\frac{\alpha}{r(1-r)}\\ &=\frac{1}{r(1-rh)}+\alpha\left[\frac{1}{r(1-r)}-\frac{h}{1-rh}\right].\end{aligned}$$

Since $rh<1/2$,

$$\frac{1}{r(1-r)}-\frac{h}{1-rh}=\frac{1-2rh+r^2h}{r(1-r)(1-rh)}>0,$$

and from $0 \leqq \alpha \leqq 1$, we have

$$\begin{aligned}\frac{|P|}{\mathrm{OPT}(L)}=\frac{H_S+H_D}{\mathrm{OPT}(L)} &\leqq \frac{1}{r(1-rh)}+\frac{1}{r(1-r)}-\frac{h}{1-rh}\\ &=\frac{1}{r}+\frac{1}{r(1-r)}.\end{aligned}$$

It remains to show that the bound of $1/r+1/r(1-r)$ is tight. Let t and u be positive integers, and let k be a positive integer such that $r^k<1-r$. Let $0<\varepsilon \ll 1$. The list L consists of two groups of squares. The first group has t squares $s_1, \cdots, s_t$, where s_i has size $r^{k+i-1}+\varepsilon$. The second group contains $\lceil r^{-k-2t-2u}\rceil$ squares of size $r^{k+t+u}+\varepsilon$.

Let P be an NFS_r or FFS_r packing of L. Each square s_i of the first group is packed into a shelf of height r^{k+i-2} in P. Let $s=\lfloor 1/(r^{k+t+u}+\varepsilon)\rfloor$. Exactly s squares of the second group are packed into each shelf of height $r^{k+t+u-1}$. Let $y=\lceil\lceil r^{-k-2t-2u}\rceil/s\rceil$. Note that if ε is sufficiently small,

$$r^{-k-t-u}-2 \leqq s < r^{-k-t-u}$$

and

$$r^{-t-u} \leqq y \leqq \frac{r^{-k-2t-2u}+1}{r^{-k-t-u}-2}+1.$$

Let $|P|$ denote the height of P. Since the top square in P has its top above the bottom of the top shelf,

$$\begin{aligned}|P| \geqq \sum_{i=1}^{t} r^{k+i-2}+(y-1)(r^{k+t+u-1}) &= r^{k-1}\sum_{i=0}^{t-1} r^i+(y-1)(r^{k+t+u-1})\\ &=\frac{r^{k-1}}{1-r}-\frac{r^{k+t-1}}{1-r}+(y-1)(r^{k+t+u-1}).\end{aligned}$$

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An optimal packing of L is no worse than the following. By choice of k and sufficiently small ε, the squares of the first group can be packed in a row of width $\leqq r^k/(1-r)$ across the bottom of the bin. The squares of the second group can be packed in rows wherever they fit to the right and above the squares of the first group. For any $\delta > 0$ it is possible to pick k and u large enough to make the total width and area of the first group small enough and the squares of the second group small enough for the total height of the packing to be at most $y(r^{k+t+u}+\varepsilon)+\delta$. Thus, we have

$$\frac{|P|}{\mathrm{OPT}(L)} \geqq \frac{\dfrac{r^{k-1}}{1-r} - \dfrac{r^{k+t-1}}{1-r} + (y-1)(r^{k+t+u-1})}{y(r^{k+t+u}+\varepsilon)+\delta}$$

$$\geqq \frac{\dfrac{r^k}{r(1-r)} - \dfrac{r^{k+t-1}}{1-r}}{\left(\dfrac{r^{-k-2t-2u}+1}{r^{-k-t-u}-2}+1\right)(r^{k+t+u}+\varepsilon)+\delta} + \frac{(y-1)(r^{k+t+u-1})}{y(r^{k+t+u}+\varepsilon)+\delta}.$$

For any $n > 0$ and $\delta' > 0$, it is possible to pick sufficiently small δ and ε and sufficiently large k, t and u such that $\mathrm{OPT}(L) > n$ and

$$\frac{|P|}{\mathrm{OPT}(L)} \geqq \frac{1}{r(1-r)} + \frac{1}{r} - \delta'.$$

Therefore, the asymptotic bound of $1/r(1-r)+1/r$ is tight. □

COROLLARY 3.1. *The value of r, $0<r<1$, for which the absolute worst case performance bound of* NFS_r *and* FFS_r *is minimal on lists of squares is* $2+\sqrt{2}$. *At this value of r,* NFS_r, $\mathrm{FFS}_r \approx 5.83\ \mathrm{OPT}(L)$ *for every list L of squares.*

The above theorem shows that the absolute worst case performance bounds of NFS_r and FFS_r improve from $2/r+1/r(1-r)$ or $1.7/r+1/r(1-r)$, respectively, to $1/r+1/r(1-r)$ for lists of squares. We do not have tight asymptotic worst case bounds of FFS_r and NFS_r on lists of squares. However, we will prove a number of lower bound results that are summarized in Tables 1 and 2. These results suggest that the asymptotic bounds are not in general much better for lists of squares than for arbitrary lists of rectangles.

TABLE 1

Lower bounds on the asymptotic worst case bound of NFS_r *on lists of squares for various values of r. Note that the asymptotic worst case bound for* NFS_r *on arbitrary lists of rectangles is* $2/r$.

r	result	squares LB
$r>1/2$, root of $1/2$	Theorems 4, 6	$1.6/r$, 1.7
$r>1/2$, not a root of $1/2$	Theorem 5	2.0
$r=1/2$	Theorem 4	3.2
$r<1/2$	Proposition 1	$1/r>2$
root of $1/k$	Theorem 4	$(k+1)/kr$

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TABLE 2

Lower bounds on the asymptotic worst case bound of FFS_r *on lists of squares for various values of r. Note that the asymptotic worst case bound for* FFS_r *on arbitrary lists of rectangles is* $1.7/r$.

r	result	squares LB
root of $1/3$ or $1/6$	Theorem 6	1.69
not a root of $1/3$ or $1/6$	Theorem 6	1.7
root of $1/2$	Theorem 4	$1.6/r$
root of $1/3$	Theorem 4	$1.36/r$
root of $1/k$	Theorem 4	$(k+1)/k$
not a root of $1/k$, for any integer $k>0$	Proposition 1	$1/r$

THEOREM 4. *Let k be a positive integer, and let r be a positive root of* $1/k$. *Let* α *and* β *be constants such that for every list L of squares of size at most H*

$$\mathrm{FFS}_r\,(L) \leqq \alpha\ \mathrm{OPT}\,(L) + \beta H$$

or for every list L of squares of size at most H

$$\mathrm{NFS}_r\,(L) \leqq \alpha\ \mathrm{OPT}\,(L) + \beta H.$$

Then

$$\alpha \geqq \frac{1}{r}\sum_{i=1}^{\infty}\frac{k-1}{k^i-1} > \frac{k+1}{kr}.$$

Proof. Let $0<\varepsilon \ll 1$. For integers j, $n>0$, the list L_{nj} will contain $(k-1)k^i n$ squares of size $1/k^i+\varepsilon$ for $1 \leqq i \leqq j$. If ε is sufficiently small, NFS_r or FFS_r will pack the squares of size $1/k^i+\varepsilon$ into $\lceil (k-1)k^i n/(k^i-1)\rceil$ shelves of height $1/k^i r$. Thus, the total height $|P|$ of the packing P is at least

$$\frac{1}{r}\sum_{i=1}^{j}\left\lceil\frac{(k-1)k^i n}{k^i-1}\right\rceil\frac{1}{k^i} - \frac{1}{k_j r} + \frac{1}{k^j} + \varepsilon \geqq \frac{n}{r}\sum_{i=1}^{j-1}\frac{1}{k^i-1}.$$

For sufficiently small ε, $(k-1)\sum_{i=1}^{j}(1/k^i+\varepsilon)<1$, and an optimal packing of L_{nj} is no worse than the following. For $1 \leqq i \leqq j$, pack the $(k-1)k^i n$ squares of size $1/k^i+\varepsilon$ in $k-1$ columns of equal height, with all of the $j(k-1)$ columns side by side. The total height of the packing is at most

$$\max_{1\leqq i\leqq j} k^i n\left(\frac{1}{k^i}+\varepsilon\right) = n + k^j n\varepsilon.$$

Thus,

$$\frac{|P|}{\mathrm{OPT}\,(L)} \geqq \frac{\dfrac{n}{r}\displaystyle\sum_{i=1}^{k}\frac{k-1}{k^i-1}}{n+k^j n\varepsilon}.$$

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For any n and any $\delta > 0$, j can be chosen sufficiently large and ε can be chosen sufficiently small such that

$$\frac{|P|}{\mathrm{OPT}(L)} \geqq \frac{1}{r} \sum_{i=1}^{\infty} \frac{k-1}{k^i - 1} - \delta.$$

Therefore,

$$\alpha \geqq \frac{1}{r} \sum_{i=1}^{\infty} \frac{k-1}{k^i - 1}.$$

Letting

$$f_1(k) = \sum_{i=0}^{\infty} \frac{1}{k^i}$$

and

$$f_2(k) = \sum_{i=1}^{\infty} \frac{k-1}{k^i - 1} = \sum_{i=0}^{\infty} \frac{1}{k^i + k^{i-1} + \cdots + k + 1},$$

we see that

$$f_1(k) - f_2(k) = \sum_{i=1}^{\infty} \frac{k^{i-1} + k^{i-2} + \cdots + 1}{k^i(k^i + \cdots + k + 1)} < \sum_{i=1}^{\infty} \frac{1}{k^{i+1}} = \frac{1}{k^2} f_1(k).$$

Consequently,

$$f_2(k) > \left[\frac{k^2 - 1}{k^2}\right] f_1(k) = \left[\frac{k^2 - 1}{k^2}\right]\left[\frac{k}{k-1}\right] = \frac{k+1}{k},$$

and

$$\alpha \geqq \frac{1}{r} f_2(k) > \frac{k+1}{kr}. \qquad \square$$

For $k = 1/2$, $1/3$ and $1/4$,

$$\frac{1}{r} \sum_{i=1}^{\infty} \frac{k-1}{k^i - 1}$$

evaluates to more than $1.6/r$, $1.36/r$ and $1.26/r$, respectively. Since $1.6/r$ is very close to $1.7/r$, which is the asymptotic worst case bound of FFS_r on rectangles, it appears that FFS_r does not perform significantly better in the worst case for squares than for rectangles.

For values of r that are not roots of $1/k$ for some integer k, the following proposition gives a simple lower bound.

PROPOSITION 1. *Let $0 < r < 1$, and let α and β be constants such that for every list L of squares of size at most H,*

$$\mathrm{FFS}_r(L) \leqq \alpha\ \mathrm{OPT}(L) + \beta H$$

or for every list L of squares of size at most H

$$\mathrm{NFS}_r(L) \leqq \alpha\ \mathrm{OPT}(L) + \beta H.$$

Then $\alpha \geqq 1/r$.

Proof. For $\varepsilon > 0$ and any integer $n > 0$, consider a list L of $n\lfloor 1/(r+\varepsilon)\rfloor$ squares of size $r+\varepsilon$. An NFS_r or FFS_r packing P uses height at least $n-1+r+\varepsilon$, while an optimal packing uses height at most $n(r+\varepsilon)$. Thus, for any $\delta > 0$ and for sufficiently large n and sufficiently small ε,

$$\frac{|P|}{\mathrm{OPT}(L)} \geqq \frac{n-1+r+\varepsilon}{n(r+\varepsilon)} > \frac{1}{r} - \delta. \qquad \square$$

The next two theorems are different from the last two results in that the lower bounds are constants rather than constants divided by r. Tables 1 and 2 show that these theorems are weaker for some values of r than the above, but stronger for other values of r.

THEOREM 5. *Let $0 < r < 1$, where r is not a root of $1/2$. Let α and β be constants such that for any list L of squares of size at most H,*

$$\mathrm{NFS}_r(L) \leqq \alpha\ \mathrm{OPT}(L) + \beta H.$$

Then $\alpha \geqq 2$.

Proof. Let $0 < \varepsilon \ll 1$. Let L_n consist of $2n$ squares $s_1, s_2, \cdots, s_{2n}$ such that for $j \geqq 1$, s_{2j-1} has size $1/2 - j\varepsilon$ and s_{2j} has size $1/2 + (j+2)\varepsilon$. For sufficiently small ε, all of these squares will be packed in shelves of the same height h. Note that $s_{2j-1} + s_{2j} = 1 - j\varepsilon + (j+2)\varepsilon > 1$ and $s_{2j} + s_{2j+1} = s_{2j} + s_{2(j+1)-1} = 1/2 + (j+2)\varepsilon + 1/2 - (j+1)\varepsilon > 1$. Therefore, NFS_r packs each square alone in a shelf of height $h > 1/2$ and $\mathrm{NFS}_r(L) \geqq n$.

An optimal packing is no worse than the following. Pack s_1 and s_3 next to each other, then for $1 \leqq j < n-2$ pack the pairs s_{2j} and s_{2j+3} next to each other and finally pack s_{2n-2} and s_{2n} on top. This packing uses a height of $(n+1)/2 + O(n\varepsilon)$.

For any $\delta > 0$, sufficiently large n and sufficiently small ε can be selected such that

$$\frac{\mathrm{NFS}_r}{\mathrm{OPT}(L)} > \frac{n/r}{\dfrac{(n+1)}{2} + O(n\varepsilon)} > 2 - \delta. \qquad \square$$

The next result shows that the asymptotic bounds for NFS_r and FFS_r for lists of squares are at least 1.6 for all r and at least 1.7 for values of r other than roots of $1/3$ or $1/6$. Note that 1.7 is the limit of the asymptotic bound $1.7/r$ of FFS_r as $r \to 1$.

Define

$$t_1 = 2, \quad t_2 = 3, \quad t_{i+1} = t_i(t_i - 1) + 1 \quad \text{for } i \geqq 2$$

and

$$a_i = t_i - 1 \quad \text{for } i \geqq 1.$$

Define

$$\gamma = \sum_{i=1}^{\infty} \frac{1}{a_i}.$$

The above sequence has been studied in [3], [6], [11] and is closely related to sequences studied in [1].

THEOREM 6. *Let $0 < r < 1$, and let α and β be such that for every list L of squares of size at most H,*

$$\mathrm{FFS}_r(L) \leqq \alpha\ \mathrm{OPT}(L) + \beta H$$

or for every list L of squares of size at most H,

$$\mathrm{NFS}_r(L) \leqq \alpha\ \mathrm{OPT}(L) + \beta H.$$

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If r is not a root of 1/3 *or* 1/6, *then* $\alpha \geqq 1.7$. *If r is a root of* 1/3 *or* 1/6, *then* $\alpha \geqq \gamma > 1.69$.

Proof. First, we show that for every r, $0<r<1$, $\alpha > \gamma$. Then we show that if r is not a root of 1/3 or 1/6, $\alpha \geqq 1.7$.

First, suppose $r<2/3$. Let $0<\varepsilon \ll 1$. For $n \geqq 1$, let L_n be a list containing n squares of size 2/3 and $\lfloor 1/3\varepsilon \rfloor \lfloor 2n/3\varepsilon \rfloor$ squares of size ε. Note that $\mathrm{OPT}(L_n)=2n/3$. Since FFS_r and NFS_r pack the large squares into shelves of height 1 and the small squares into shelves of total height at least $\lfloor 1/3\varepsilon \rfloor \lfloor 2n/3\varepsilon \rfloor \varepsilon^2 \geqq (1/3\varepsilon - 1)(2n/3\varepsilon - 1)\varepsilon^2 = 2n/9 - O(\varepsilon) + O(\varepsilon^2)$,

$$\mathrm{FFS}_r(L_n) = \mathrm{NFS}_r(L_n) > \tfrac{11}{9}n - O(\varepsilon) - O(\varepsilon^2).$$

Thus, for any δ, ε may be packed sufficiently small such that

$$\frac{\mathrm{FFS}_r(L_n)}{\mathrm{OPT}(L_n)} = \frac{\mathrm{NFS}_r(L_n)}{\mathrm{OPT}(L_n)} > \left(\frac{11}{9}\cdot\frac{3}{2}\right) - \delta > \gamma - \delta.$$

Since this is true for every $n>0$, FFS_r and NFS_r cannot have an asymptotic bound $<\gamma$.

Now, suppose $r \geqq 2/3$. Let $\varepsilon \ll 1$. For n, $s \geqq 1$, define $L_{n,s}$ to be a list containing nt_i squares of size $1/t_i + \varepsilon/i$, $1 \leqq i \leqq s$. It may be easily verified that $\sum_{i=1}^{\infty} 1/t_i = 1$. Thus, for sufficiently small ε, $L_{n,s}$ size $1/t_i + \varepsilon/i$, $1 \leqq i \leqq s$. It may be easily verified that $\sum_{i=1}^{\infty} 1/t_i = 1$. Thus, for sufficiently small ε, $L_{n,s}$ can be packed into s columns, such that the ith column contains the squares of size $1/t_i + \varepsilon/i$. Thus, $\mathrm{OPT}(L_{n,s}) \leqq n + nt_s/s\varepsilon$.

Now, consider a NFS_r or FFS_r packing of $L_{n,s}$. For $i \geqq 1$ and sufficiently small ε,

$$\frac{\dfrac{1}{t_{i+1}} + \dfrac{\varepsilon}{i+1}}{\dfrac{1}{t_i} + \dfrac{\varepsilon}{i}} < \frac{\dfrac{1}{t_2} + \dfrac{\varepsilon}{2}}{\dfrac{1}{t_1} + \varepsilon} = \frac{\dfrac{1}{3} + \dfrac{\varepsilon}{2}}{\dfrac{1}{2} + \varepsilon} < \frac{2}{3}.$$

Therefore, squares of distinct sizes are packed into shelves of distinct heights. Since only $t_i - 1$ squares of size $1/t_i + \varepsilon/i$ can fit into a shelf,

$$\begin{aligned}\mathrm{NFS}_r(L_{n,s}) = \mathrm{FFS}_r(L_{n,s}) &\geqq \sum_{i=1}^{s} \frac{nt_i}{t_i - 1}\left(\frac{1}{t_i} + \frac{\varepsilon}{i}\right)\\ &= n\sum_{i=1}^{s}\frac{1}{t_i - 1} + n\sum_{i=1}^{s}\frac{t_i\varepsilon}{i(t_i-1)} > n\sum_{i=1}^{s}\frac{1}{a_i}.\end{aligned}$$

Consequently, for any $\delta > 0$, there are a sufficiently small ε and sufficiently large s such that

$$\frac{\mathrm{NFS}_r(L)}{\mathrm{OPT}(L)} = \frac{\mathrm{FFS}_r(L)}{\mathrm{OPT}(L)} > \frac{n\sum_{i=1}^{s}\dfrac{1}{t_i}}{n(1+\varepsilon)} > \gamma - \delta.$$

Since this is true for any n, FFS_r and NFS_r cannot have an asymptotic bound smaller than γ.

Now, consider any r which is not a root of 1/3 or 1/6. We modify the list used in the proof that the asymptotic bound for FF is at least 1.7 [11]. Let n be a positive integer divisible by 17 and let $0<\delta \ll 18^{-6n/17}$. For $1 \leqq i \leqq 6n/17$, define $\delta_i = 18^{(6n/17)-i}\delta$. For $1 \leqq i \leqq 6n/17$, define

$$\begin{aligned} a_{0i} &= 1/6 + 33\delta_i, & a_{4i} &= 1/6 - 13\delta_i,\\ a_{1i} &= 1/6 - 3\delta_i, & a_{5i} &= 1/6 + 9\delta_i,\\ a_{2i} = a_{3i} &= 1/6 - 7\delta_i, & a_{6i} = a_{7i} = a_{8i} = a_{9i} &= 1/6 - 2\delta_i.\end{aligned}$$

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The first part of L will consist of squares of size $a_{01}, a_{11}, \cdots, a_{91}, a_{02}, \cdots, a_{92}, \cdots, a_{0(6n/17)}, \cdots, a_{9(6n/17)}$. Note that for sufficiently small δ, FFS$_r$ will pack all of these squares into shelves of the same height, since r is not a root of $1/6$. Also, note that $\sum_{j=0}^{4} a_{ji} = 5/6+3\delta_i$ and $\sum_{j=5}^{9} a_{ji} = 5/6+\delta_i$. Since $5/6+\delta_i = 5/6+18\delta_{i+1}$, a_{jk} will not fit into a shelf containing $\sum_{j=0}^{4} a_{ji}$ or $\sum_{j=5}^{9} a_{ji}$ for $k>i$. Therefore, for $1 \leqq i \leqq 6n/17$, $a_{1i}, \cdots, a_{4i}$ are packed by FFS$_r$ into shelf $2i-1$ and $a_{5i}, \cdots, a_{9i}$ are packed into shelf $2i$. Thus, the total height used for these shelves is at least $2n/17$.

For $1 \leqq i \leqq 3n/17$, define

$$b_{0i} = 1/3+46\delta_{2i}, \qquad b_{4i} = 1/3+12\delta_{2i},$$
$$b_{1i} = 1/3-34\delta_{2i}, \qquad b_{5i} = 1/3-10\delta_{2i},$$
$$b_{2i} = b_{3i} = 1/3+6\delta_{2i}, \qquad b_{6i} = b_{7i} = b_{8i} = b_{9i} = 1/3+\delta_{2i}.$$

Note that for each i, $b_{0i}+b_{1i} = b_{2i}+b_{3i} = 2/3+12\delta_{2i}$ and $b_{4i}+b_{5i} = b_{6i}+b_{7i} = b_{8i}+b_{9i} = 2/3+2\delta_{2i}$. For $k>i$, $2/3+2\delta_{2i} > 2/3+36\delta_{2k}$ and b_{jk} cannot fit into a shelf already containing b_{ji}, $b_{(j+1)i}$ for j even. Also, b_{jk} cannot be packed into a shelf containing five a_{rs}'s, for $r, s \geqq 0$. Therefore, FFS$_r$ packs each group $b_{0i}, \cdots, b_{9i}$ into 5 successive bins, using a total height of at least $5(3n/17)(1/3) = 5n/17$.

The remainder of the list consists of $20n/17$ squares of size $1/2+\delta$. The shelves for these squares use height at least $10n/17$.

The total height used by the FFS$_r$ packing is thus at least $2n/17+5n/17+10n/17 = n$.

An optimal packing is no worse than the following. Pack the squares of size $1/2+\delta$ in a column at the left side of the bin. The remaining squares will be packed next to these n groups of three, with a pair of squares of size about $1/6$ packed one above the other next to a square of size about $1/3$. The groups of three are as follows:

(a) $a_{j(2i-1)}, a_{j(2i)}, b_{ji}, j = 2, 3, 4, 6, 7, 8, 9, 1 \leqq i \leqq 3n/17$,
(b) $a_{5(2i-1)}, a_{5(2i)}, b_{5(i-1)}, 1 < i \leqq 3n/17$,
(c) $a_{0(2i-1)}, a_{0(2i)}, b_{1(i-1)}, 1 < i \leqq 3n/17$,
(d) $a_{1(2i-1)}, a_{1(2i)}, b_{0(i+1)}, 1 \leqq i < 3n/17$,
(e) $a_{1(6n/17-1)}, a_{1(6n/17)}, b_{5(3n/17)}$.

Note that each triple requires height at most $1/3+66\delta_1$, and the above $7(3n/17)+3(3n/17-1)+1$ triples use height $\leqq (30n/17-2)(1/3+66\delta_1)$. This leaves a_{51}, a_{52}, a_{01}, a_{02}, $b_{1(3n/17)}$ and b_{01}. The $b_{1(3n/17)}$ and b_{01} can be packed above each other using height at most $2(1/3+46\delta_1)$. At this point, the right side of the packing uses height

$$\begin{aligned}(30n/17-2)&(1/3+66\delta_1)+2(1/3+46\delta_1)\\ &= 10n/17+(30n/17-2)(66\delta_1)+92\delta_1\\ &= 10n/17+[(30n/17-2)66+92]\delta \cdot 18^{(6n/17)-1}\end{aligned}$$

which is greater than the height of the left side, which is $(20n/17)(1/2+\delta) = 10n/17+(20n/17)\delta$. The remaining 4 a_{ij}'s can be packed above everything else using height $1/6+33\delta_1$. Thus, the total height is at most $10n/17+1/6+(30n/17)66\delta \cdot 18^{(6n/17)-1}$.

Since FFS$_r$ $(L) \geqq n$ and δ may be picked to be as small as desired, the asymptotic bound cannot be less than $17/10$. □

5. Conclusions. Theorem 2 shows that the asymptotic performance of FFS$_r$ is no worse than $(1.7/r)$ OPT $(L)+H/r(1-r)$, where H is the height of the tallest piece. Thus, we can get asymptotic performance arbitrarily close to the 1.7 bound of FFDH

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[7] without having to sort the rectangles first. Corollary 2.1 shows that the absolute worst case performance is worse than for FFDH (Golan [10] proved that NFD $(L) \leqq 3$ OPT (L) and the same proof works for FFDH). Absolute worst case bounds are a good measure of performance only for small numbers of rectangles. For large numbers of rectangles, the asymptotic bounds are a better measure. The shelf algorithms are intended primarily for situations requiring packing large numbers of rectangles, and the absolute worst case performance is not a primary consideration.

The previous papers on two-dimensional packing problems have not required that rectangles be packed in the order given; the algorithms order the lists before packing them. Since the FFS_r algorithm can pack pieces in any order, it is natural to ask whether ordering the lists would improve its worst case bounds.

In one-dimensional bin packing, the first-fit algorithm has an asymptotic bound of 17/10 if, the pieces are packed in the order given, and 11/9 if they are packed in order of decreasing size [11]. Now, the decreasing size of the one-dimensional case corresponds to decreasing width in the two-dimensional case. The one-dimensional results suggest that packing rectangles by decreasing width using the FFS algorithm might obtain asymptotic bounds better than 17/10. Unfortunately, it is easily seen that even using decreasing width, the asymptotic bound of FFS_r cannot be better than 3/2 for any r (consider, for example, lists with n pieces of height 1 and width $\varepsilon/2$ and $n\lfloor 1/\varepsilon \rfloor$ pieces of height 1/2 and width ε). Since the up-down algorithm [2] has an asymptotic bound of 5/4, such an asymptotic bound would not be an improvement over existing algorithms which reorder the pieces before packing them.

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